

Topic:

Web Services

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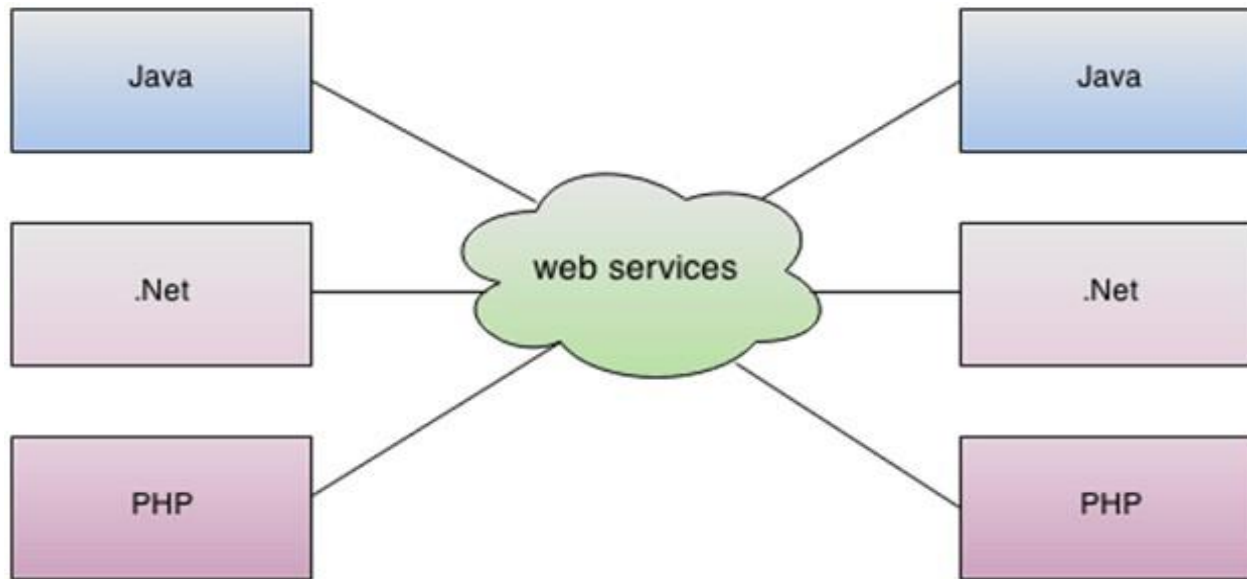
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INTRODUCTION TO WEB SERVICES

- *It is a client-server application or application component for communication.*

- *The method of communication between two devices over the network.*
- *It is a software system for the interoperable machine to machine communication.*
 - *It is a collection of standards or protocols for exchanging information between two devices or application.*

Let's understand it by the figure given below:

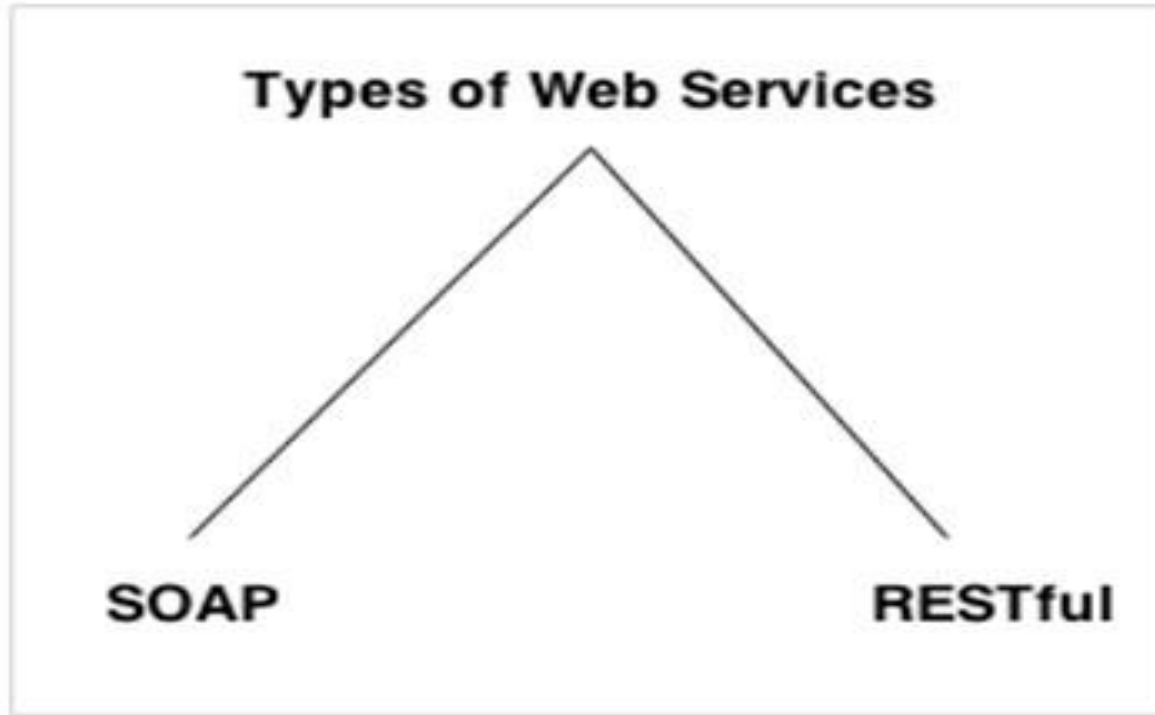


Types of Web Services

There are mainly two types of web services.

1. SOAP API

2. *RESTful API.*



1. SOAP api:

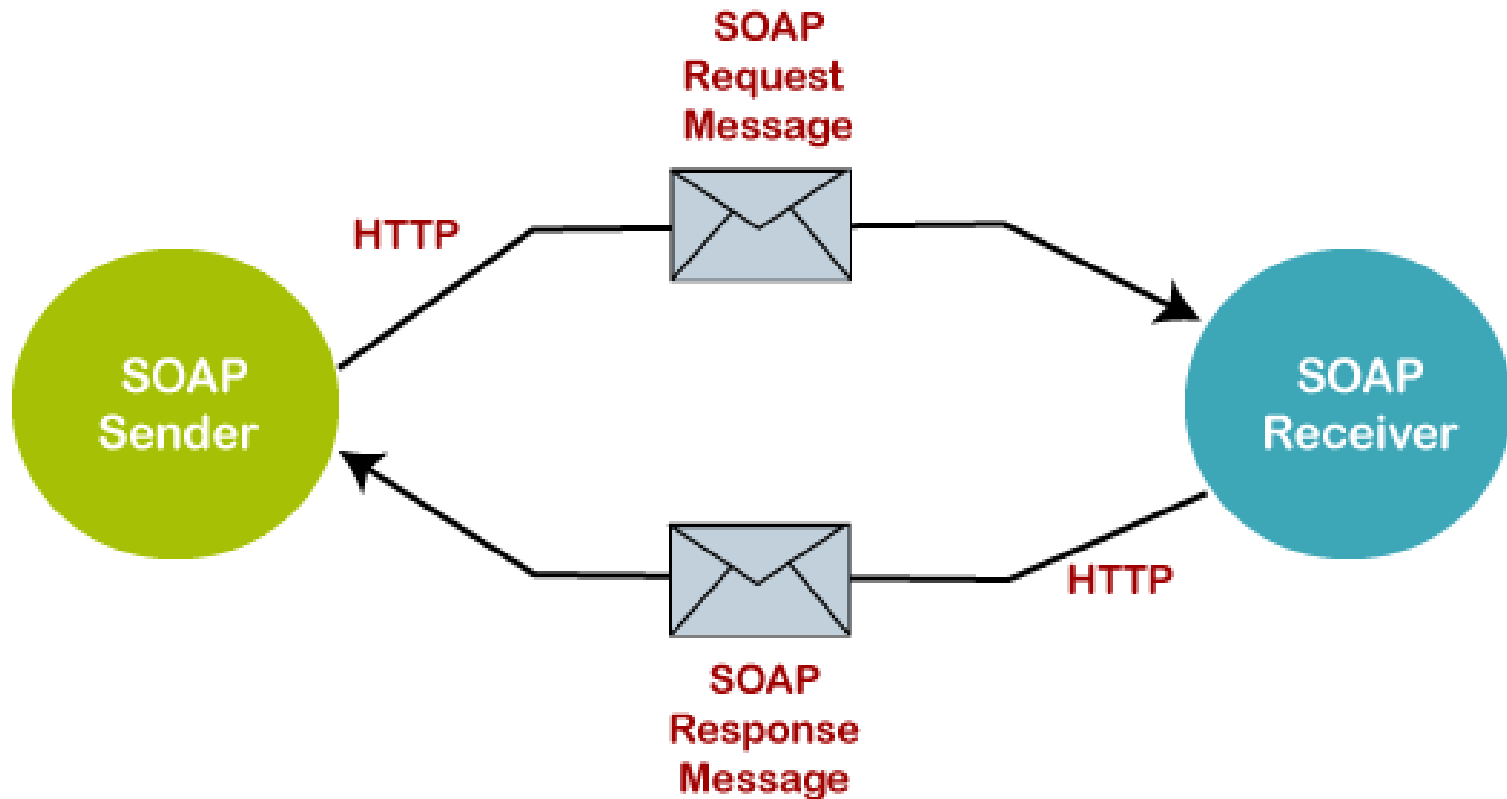
- SOAP is an important protocol that helped introduce the widespread use of Web Services, also called APIs. Based on XML, the SOAP protocol is still in wide usage.

- Many organizations use the more flexible REST API pattern but others prefer the structure, datatype control, and defined standard of SOAP.
- SOAP is the Simple Object Access Protocol, a messaging standard defined by the World Wide Web Consortium and its member editors.

EXAMPLE OF SOAP API:

- This ISBN validation service uses a standardized SOAP envelope • to pass a structured message as part of the request, resulting
- in a standardized response sent in the same way.
- The SOAP response structure makes it easy for developers to
- understand and put to work in their applications and integrations.

DIAGRAM OF SOAP API

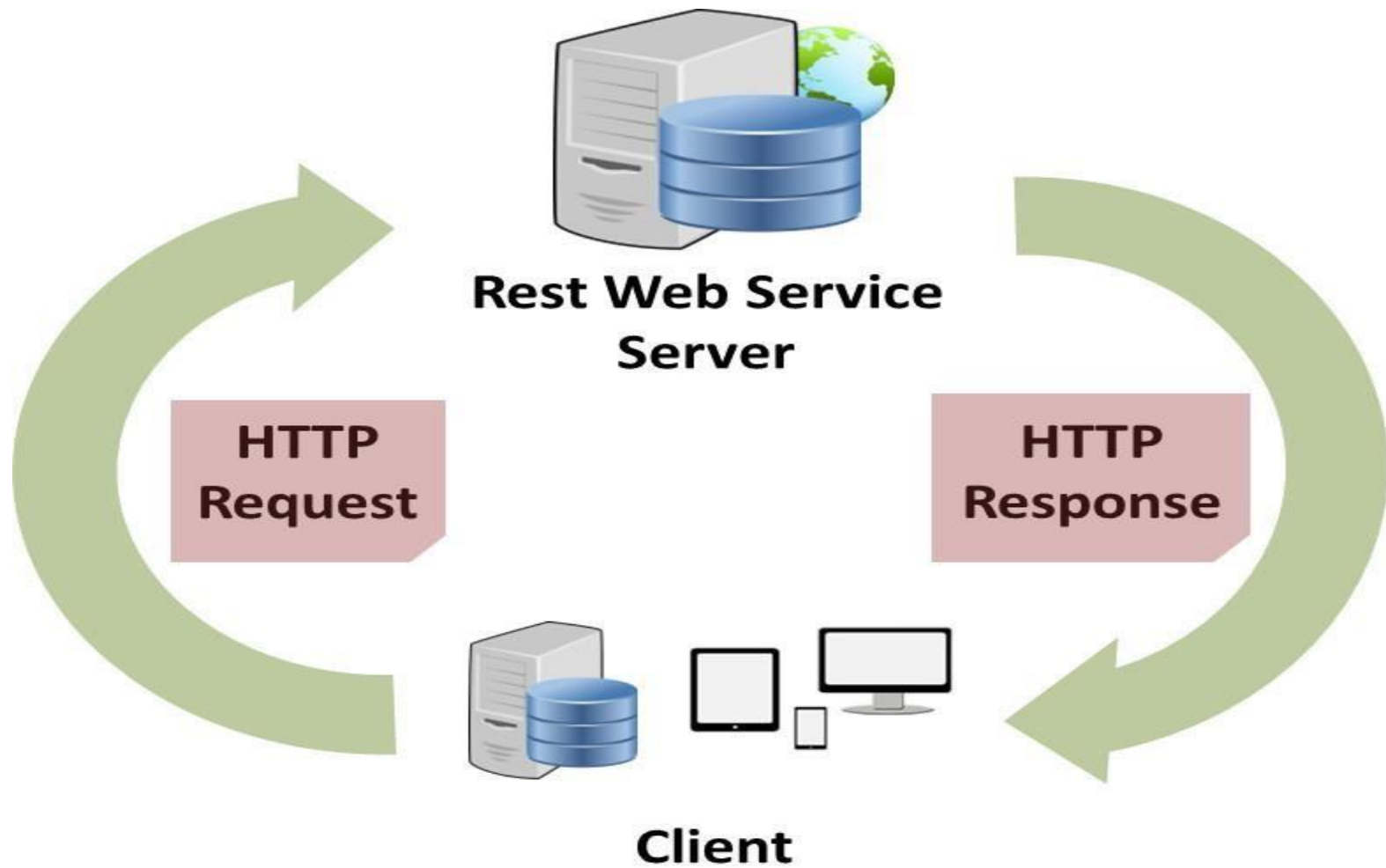


RESTFULL WEB SERVICES

- REST APIs serve as the middleman between your client and your host server, fielding resource requests and delivering responses via HTTP.

- In a REST design, the client initiates the interaction — for example, accessing a webpage via a URL is a commonly used HTTP request.

DIAGRAM OF RESTFUL WEBSERVICES:



PRIVACY:

- Privacy is the ability of an individual or group to keep away themselves (or information Related to them) from other people or group.
- A state in which one is not observed or disturbed by other people.
- Data privacy is the protection of personal data from those who should not have access to it and the ability of individuals to determine who can access their personal information.

DIGITAL PRIVACY

Digital privacy refers to the protection of individual's information that is Used or created while using the internet on the computer or the personal Devices .

- have better control over the personal information.

- Individual should be able to choose what space within their personal domain can be accessed by others and to control the manner ,scope and timings of its Exposure.

DIGITAL PRIVACY PRINCIPLES

Following are the principles of digital privacy

- **Data minimization:**

The principle of “data minimisation” means that a data controller should limit the collection of personal information to what is directly relevant and necessary to accomplish a specified purpose. They should also retain the data only for as long as is necessary to fulfil that purpose.

- **Transparency:**

- Transparency is a core principle in data protection. You have the right to know which of your personal data are collected, used, consulted or otherwise processed and to what extent the personal data are or will be processed.
- The principle of transparency requires that any information or communication relating to the processing of personal data is easily accessible and easy to understand, and that clear and plain language be used.
- **Accuracy:**
- Controllers must ensure that personal data are accurate and, where necessary, kept up to date; taking every reasonable step to ensure that personal data that are inaccurate, having regard to the purposes for which they are processed, are erased or rectified without delay.

- The principle of accuracy of personal data states that reasonable effort should be made to ensure that the data of an individual is accurate and kept up to date
- **Data security**
- Data security protects information from unauthorized access,
 - use, and disclosure.
- It also protects it from disruption, modification, or destruction.
- Data privacy is the right to control who gets to see your personal information like credit card numbers and bank account balances.

GDPR

- The General Data Protection Regulation (GDPR) is a legal framework that sets guidelines for the collection and processing

of personal information from individuals who live and outside of the European Union (EU).

- The EU general data protection regulation (GDPR) governs how the personal data of individuals in the EU may be processed and transferred.

PRINCIPLES OF GDPR

Following are the seven principles of gdpr

- Lawfulness, fairness and transparency.
- Purpose limitation.
- Data minimisation.
- Accuracy.
- Storage limitation.
- Integrity and confidentiality (security)
- Accountability.

• **P3P**

- Privacy is all about keeping your personal information safe and secure, like your name, address, or things you do online. P3P stands for Platform for Privacy Preferences. It's a way for websites to tell you how they handle your info. It's like a privacy agreement between you and the website!
- **Important information in privacy policies:**
- Different users are looking for different things, but many common element What information will be collected about me?
- How will it be used?
- Will it be shared with other companies Will it be used to send me unsolicite marketing?

- How can I opt-out? (but many users not aware this is possible)

- **The companies have adopted two different approaches in this regard:**

1.opt-out approach (most commonly used)

2.opt-in approach (Less commonly used)

- The opt-out approach means you're automatically included in something unless you say no. On the other hand, the opt-in approach means you have to actively choose to be included. It's like being added to a mailing list without asking (opt-out) versus signing up for it yourself (opt-in). It's all about whether you have to take action to be part of something or not!

- **guiding principles to form the basis of any privacy legislation:**

1. Collected data may be used for improved customer service
2. Sharing of personal data with outside firms/persons should not be allowed unless the customer consents to that
3. Customers should have the right to receive information about what type of data has been collected from them and in what manner has it been used;
4. Customers should have the right to ask for the deletion of any of their data collected by the company.

• **P3P Definition**

- P3P is a system for making Web site privacy policies machine-readable P3P enhances user control by putting privacy policies where users can find them, in a form users can understand, and enables users to act on what they see. (e.g., a popup)²⁴

P3P was developed by the World Wide Web Consortium (W3C). Microsoft Internet Explorer is the only major browser to support P3P. Companies use this information to provide specifically targeted advertisements.

• **Goals of P3P:**

According to the W3C the main goal of P3P is to increase user trust and confidence in the Web through technical empowerment.

P3P manages information through privacy policies

.When a website uses P3P, they set up a set of policies

• **P3P and its Working:**

When a user decides to use P3P, they set their own set of policies and state what personal information they will allow to be seen by the sites that they visit

Then when a user visits a site, P3P will compare what personal information the user is willing to release, and what information the server wants to get if the two do not match, P3P will inform the user

and ask if he/she is willing to proceed to the site, and risk giving up more personal information

The privacy policy can be retrieved as an XML file or can be included, in compact form, in the HTTP header

The location of the XML policy file that applies to a given document can be

specified in the HTTP header of the documents

specified in the HTML head of the document

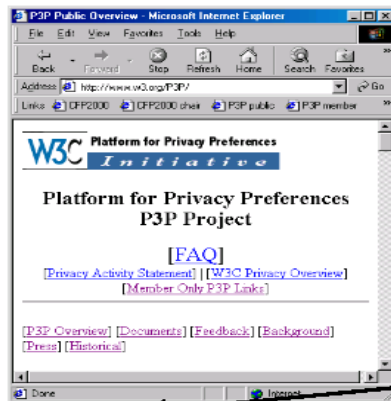
What's in a P3P policy?

- Name and contact information for site
- The kind of access provided
- Mechanisms for resolving privacy disputes
- The kinds of data collected ■ How collected data is used, and whether individuals can opt-in or opt-out of any of these uses

■ Whether/when data may be shared and whether there is opt-in or opt-outData retention policy

- **Check P3P POLICY:**

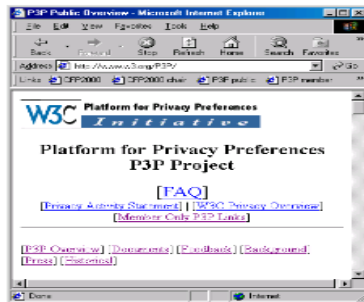
- To check a website's P3P policy, you can look for a privacy policy link on the website.
- Sometimes, websites mention their privacy practices in the footer section or under the settings menu.
- By clicking on the privacy policy link, you can usually find information about how the website handles your data and if they follow the P3P standards
- It's a good way to know how your info is being taken care of online!



GET /index.html HTTP/1.1
Host: www.att.com
. . . Request web page

HTTP/1.1 200 OK
Content-Type: text/html
. . . Send web page

Web
Server



GET /w3c/p3p.xml HTTP/1.1
Host: www.att.com
Request Policy Reference File

Send Policy Reference File

Request P3P Policy

Send P3P Policy

GET /index.html HTTP/1.1
Host: www.att.com
. . . Request web page

HTTP/1.1 200 OK
Content-Type: text/html
. . . Send web page

Web
Server

• **Advantages:**

1. Helps users understand how their personal information is being used
2. Provides transparency and control over data collection.
3. Can enhance user trust in websites that adhere to P3P standards
4. . Simplifies the process of reviewing privacy policies for users.
5. Encourages websites to clearly outline . their data practices
6. Promotes consistency in privacy practices . across websites.
7. Can lead to improved data protection and privacy for users.

• **Disadvantages:**

1. Some websites may manipulate or misrepresent their P3P policies.
2. Users may not fully understand the technical language used in policies.
3. . P3P may not cover all aspects of data privacy and security.
4. . Compliance with P3P standards can be challenging for some websites.
5. Lack of enforcement mechanisms to ensure accurate P3P disclosures.
6. P3P policies may not keep up with rapidly changing technology and data practices.

7. Users may become complacent assuming all websites with P3P are trustworthy.